EFU AVIATION LICENCE v1.0

Aviation & Airspace Sovereignty Audit Standard
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0. Purpose and Scope

Experimental **EFU-based licensing and measurement framework for commercial aviation** (passenger, cargo, business flights).

NOT a replacement for: EASA operating permits, EU ETS aviation quotas, or CORSIA, but an **open research protocol** measuring aviation's **physical footprint** in human-scale **EFU metrics**:

- Kerosene (Jet A-1)
- Carbon + contrail forcing
- Noise/entropy

I. Core Concepts

EFU-Air (EFU-AIR)

```text
EFU-H_AIR: Fuel consumption (Jet A-1 kerosene MWh)
EFU-C_AIR: CO₂ + contrail forcing (high-altitude effects)
EFU-S_AIR: Noise/entropy + spacetime distortion equivalent
EFU-M_AIR: Maintenance (spare parts, paint, lubricants)
EFU-W_AIR: Water vapor condensation (contrails, cirrus clouds)
```

Aviation Sovereignty Gap (SS_AIR)

```text
SS_AIR = TEFU_AIR_total – Regional Airspace Carrying Capacity (RACC_AIR)
```

Measures if flight traffic stays within airspace thermodynamic regeneration capacity.

MROI_AIR

```text
MROI_AIR = THI_AIR / TEFU_AIR
THI_AIR = economic connectivity + temporary mobility + cargo efficiency
```

II. EFU Aviation Licence – Research Status

Documents EFU-audited airlines, airports, flight routes.
Does NOT replace safety certifications, air traffic quotas, or fuel taxes.

Enables airlines and airports benchmarking in EFU context.

III. Metrics and Thresholds

1. EFU Components (per passenger-km or tonne-km)

EFU-H_AIR (Fuel)

```text
EFU-H_AIR = kerosene_MWh / 12.88 / passenger_km
Example: Economy short-haul = 0.04 liter/100km/pax → 0.00012 EFU/pax-km
```

EFU-C_AIR (Carbon + contrail forcing)

```text
EFU-C_AIR = CO2e_kg / 4t/person/year / passenger_km
RFI factor: 2-3× high-altitude multiplier
```

EFU-S_AIR (Noise/Entropy)

```text
EFU-S_AIR = noise_dB_equivalent / 11500 / passenger_km
Experimental: flight turns entropy + atmospheric mixing
```

TEFU_AIR

```text
TEFU_AIR = 1.2×EFU-H + 2.0×EFU-C + 1.5×EFU-S + 0.5×EFU-M + 0.8×EFU-W
```

2. Regional Airspace Carrying Capacity (RACC_AIR)

```text
RACC_AIR = Sustainable_flight_traffic_billion_pax_km / Population / 2,000 pax_km/person/year
Example: EU = 1,000B pax-km / 448M / 2,000 = 1.12 EFU_AIR/person/year
```

IV. Licence Categories (EFU-AIR Labels)

Level	SS_AIR	MROI_AIR	Label	Interpretation
1	≤ 0	≥1e-3	**EFU-AIR SOVEREIGN**	Within airspace capacity

Level	SS_AIR	MROI_AIR	Label	Interpretation
2	0 < ≤20k EFU	≥5e-4	**EFU-AIR BALANCED**	Monitorable
3	20k < ≤100k EFU	≥1e-4	**EFU-AIR DEFICIT**	Airspace overload
4	>100k EFU	<1e-4	**EFU-AIR CRITICAL**	Thermodynamic limit

V. Regional RACC_AIR Calibrations

Region	Flight Traffic (B pax-km/year)	Population (M)	RACC_AIR (EFU/person/year)
EU	1,000	448	1.12
USA	1,500	340	2.20
CHINA	800	1,410	0.28
JAPAN	200	125	1.28
WORLD	5,000	8,000	0.31

VI. Example: Ryanair Short-Haul Fleet (EU)

```text
Input (annual operations):
Passengers: 150M
Average distance: 1,200 km
Kerosene: 5B liters (18,000,000 MWh)
Core staff: 15,000 personnel
Region: EU (RACC_AIR = 1.12 EFU/person/year)

Calculation:
EFU-H_AIR = 18M/12.88/180B = 0.000078 EFU/pax-km
EFU-C_AIR = 45M tCO2/4/180B = 0.000063 EFU/pax-km (RFI=2.5)
EFU-S_AIR = noise/entropy ≈ 0.00002 EFU/pax-km

TEFU_AIR_total ≈ 28,000,000 EFU/year
Allocated = 15,000 × 1.12 = 16,800 EFU/year
SS_AIR = 28M – 16,800 = +27,983,200 EFU/year → **CRITICAL (Level 4)**
```

VII. Implementation Notes

```text
EU ETS complement: TEFU_AIR basis for aviation carbon pricing
CORSIA alignment: EFU-C_AIR metric for contrail accounting
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Data sources: Tier 1 = fuel bills, flight logs; Tier 2 = Flightradar24